A Constructive Introduction to Quantum Field Theory

Stochastic observables, the Polchinski flow, FBSDEs

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SISSA lectures, Lecture 3 — notes: Lecture 3, §§4–10

Where we are

So far. A Euclidean field is a measure on $\mathscr{S}'(\mathscr{M})$; the GFF μ_m is its Gaussian part; local operators are functions of φ. Wick powers $:\varphi^n:$ are random distributions iff $n(d-2)<d$, and **not even tight** otherwise.

Today. Index everything by a **resolution** $t\in[0,\infty)$.

1. The GFF as the terminal value of a martingale of smooth fields; **observables** = martingales.
2. Interactions: the **Polchinski equation** and its **FBSDE**; the force is an observable.
3. The FBSDE in force coordinates ⇒ the **Wetterich equation**.
4. Variational structure: Boué–Dupuis, HJB, infinite volume.
5. Additional material: products and OPE, flatness, counterterms for Φ^4_3, DS equations at every scale.

The idea

- Wilson–Kadanoff: physics is organised **scale by scale**. The measure-on-$\mathscr{S}'$ picture hides this, and forces singular objects from the start.
- Observe the field at resolution t: a **smooth** function φ_t. Information at resolution t: $\mathscr{F}_t = \sigma(\varphi_s : s \le t)$.
- A physical quantity is not a random variable but a **consistent family of finite-resolution estimates**, i.e. a martingale.

Slogan

Observables are martingales in the filtration of finite-resolution measurements. They need not have a terminal value.

Caveat. Statements marked (formal) are exact in finite dimensions, with a terminal resolution $T < \infty$; the limits $T \to \infty$, $L \to \infty$ are the real problems. Bailleul–Chevyrev–G. (in prep.); Polchinski 1984; Bauerschmidt–Bodineau–Dagallier 2024

Setting: covariance profiles and observables

Lecture 3

Covariance profiles

Definition. *A **covariance profile** is a family $(Q_t)_{t\geq 0}$ of symmetric, translation invariant operators with $Q_0 = 0$, $\dot{Q}_t := \partial_t Q_t \geq 0$, $Q_t \to C = (m^2 - \Delta)^{-1}$, and smooth kernels for $t < \infty$.*

Heat profile:

$$Q_t = C\, \mathrm{e}^{-A/t^2}, \qquad \dot{Q}_t = \frac{2}{t^3}\mathrm{e}^{-A/t^2}, \qquad \dot{Q}_t(x-y) = \frac{2\, t^{d-3}\mathrm{e}^{-m^2/t^2}}{(4\pi)^{d/2}}\mathrm{e}^{-t^2|x-y|^2/4} \text{ on } \mathbb{R}^d.$$

$\dot{Q}_t$ lives on distances $\lesssim 1/t$; Q_t keeps the modes $|k| \lesssim t$: t is a **momentum scale**.

Others: sharp Fourier cutoff $1_{|k|\leq t}\,(|k|^2 + m^2)^{-1}$; **finite range** decompositions, $\dot{Q}_t(x, y) = 0$ for $|x-y| \geq 1/t$.

Notation: $c_t = Q_t(x,x)$, $Q_{T,t} = Q_T - Q_t$. For $d \geq 2$, $c_t \to \infty$ ($\frac{1}{2\pi}\log t$ in $d=2$, $\asymp t^{d-2}$ in $d \geq 3$).

The field at resolution t

W cylindrical Brownian motion on $L^2(\mathscr{M})$,

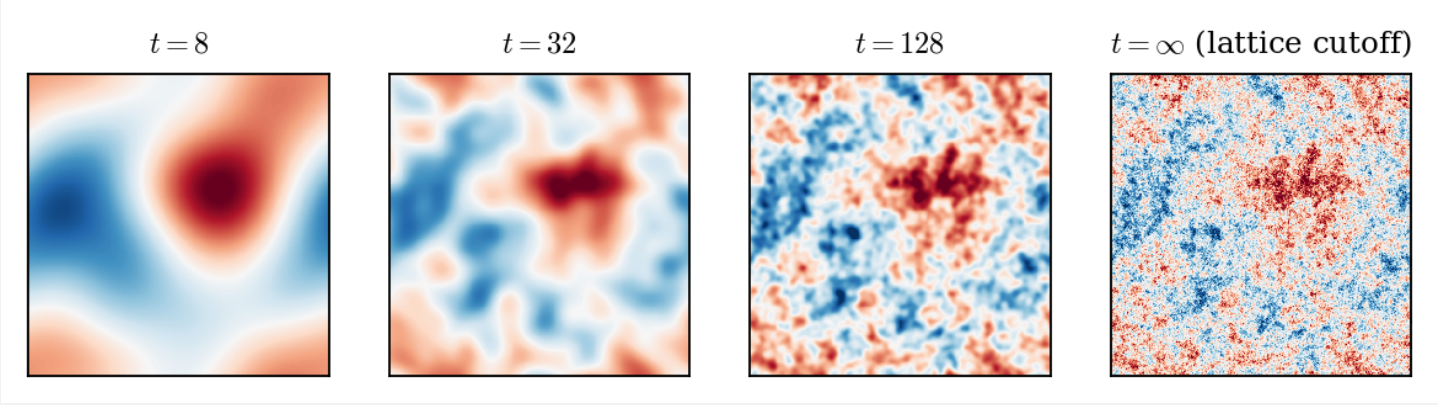
$$X_t := \int_0^t \dot{Q}_s^{1/2}\,\mathrm{d}W_s.$$

Proposition. *X_t is a smooth Gaussian field,* $\mathrm{Law}(X_t) = \mu^{Q_t}$*; $(X_t(x))_t$ is a continuous martingale with* $\mathrm{d}\langle X(x), X(y)\rangle_t = \dot{Q}_t(x,y)\,\mathrm{d}t$*; $X_t \to X_\infty$ in H^{-s}_{loc}, $s > \frac{d}{2} - 1$, with* $\mathrm{Law}(X_\infty) = \mu_m$*; $X_T - X_t \perp \mathscr{F}_t$, with law $\mu^{Q_{T,t}}$.*

The GFF is the **terminal value of a Gaussian martingale of smooth fields**.

Proof (torus). $X_t = \sum_n \left(\int_0^t \dot{q}_n(s)^{1/2}\mathrm{d}B_s^n\right) e_n$: independent Gaussian martingales, variance $q_n(t) \uparrow q_n(\infty)$; then as in Lecture 2.

The field at resolution t: a picture



Heat profile on $\mathbb{T}^2$, $m = 4$: $t = 8, 32, 128$ and the lattice cutoff, same white noise.

The martingale reveals finer and finer details; the large scales are fixed at low resolution.

Stochastic observables

$\mathbb{E}_t = \mathbb{E}[\,\cdot\,|\mathscr{F}_t]$.

Definition. (Bailleul–Chevyrev–G.) *An **observable** is an $(\mathscr{F}_t)$-martingale $O = (O_t)_{t\geq 0}$. Its expectation is $\langle O\rangle = \mathbb{E}\,O_t$. An observable field is a family $(O(x))_{x\in\mathscr{M}}$.*

- $Y \in L^1(\mathscr{F}_\infty)$ gives a **closed** observable $O_t = \mathbb{E}_t Y$.
- The point: observables need **not** be closed. $X_t(x)$, $d \geq 2$, diverges, yet is a perfectly good family of estimates.

Asymptotic uniqueness

(a) $\|O^1_t - O^2_t\|_{L^1} \to 0 \Rightarrow O^1 = O^2$. (b) N adapted, $\|N_t - O_t\|_{L^1} \to 0 \Rightarrow O_s = \lim_t \mathbb{E}_s N_t$.

Proof. $\mathbb{E}_s$ is an L^1 contraction.

Germs and locality

An observable is determined by its **asymptotics**.

- A **germ** of O: functionals $\mathring{O}_t$ of smooth fields with $\|O_t - \mathring{O}_t(X_t)\|_{L^1} \to 0$. A germ determines O.
- $O(x)$ is **local** if it has germs

$$\mathring{O}_t(X_t)(x) = \mathring{\mathcal{O}}_t(X_t(x), \nabla X_t(x), \dots, \nabla^k X_t(x)).$$

- Germs are not unique: this freedom **is** the freedom of choosing counterterms.

Example. $:X_t^n:(x) = H_n(X_t(x); c_t)$ is a local observable in **every** d (more later), although $:\varphi^n:$ does not exist as a random distribution when $n(d-2) \geq d$.

Interactions: the Polchinski equation and FBSDEs

Lecture 3

Effective potentials

Finite volume $\mathbb{T}_L^d$, terminal resolution $T < \infty$ (UV cutoff), interaction V_T bounded below, e.g.

$$V_T(\varphi) = \int_{\mathbb{T}_L^d} [\lambda H_4(\varphi; c_T) + \gamma_T H_2(\varphi; c_T)], \qquad \nu_T(\mathrm{d}\varphi) = Z_T^{-1} \mathrm{e}^{-V_T(\varphi)} \mu^{Q_T}(\mathrm{d}\varphi).$$

Definition. (Wilson effective action)

$$\mathrm{e}^{-V_t(\varphi)} := \int \mathrm{e}^{-V_T(\varphi+\zeta)} \mu^{Q_{T,t}}(\mathrm{d}\zeta), \qquad t \in [0, T].$$

The interaction felt at resolution t once the scales $[t, T]$ are integrated out; $V_0(0) = -\log Z_T$ is the free energy.

Marginals and stochastic localisation

Proposition. (Marginals) $\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{P}_0 = Z_T^{-1}\mathrm{e}^{-V_T(X_T)}$. *Then* $\mathrm{Law}_{\mathbb{P}}(X_t) = \nu_t \propto \mathrm{e}^{-V_t}\mu^{Q_t}$, *and* $\mathrm{Law}_{\mathbb{P}}(X_T|\mathscr{F}_t) = \mu_t^{X_t}$, *the tilt of* $\mu^{Q_{T,t}}(\cdot - X_t)$ *by* e^{-V_T}.

- At every resolution the field is again a **Gibbs measure**: reference Q_t, potential V_t.
- The measure-valued martingale $t \mapsto \mu_t^{X_t}$ goes from ν_T $(t=0)$ to δ_{X_T} $(t=T)$: a **stochastic localisation** of ν_T (Eldan), adapted to the covariance profile.

Proof of the marginals. Under $\mathbb{P}_0$, $X_T = X_t + (X_T - X_t)$ with independent increment $\sim\mu^{Q_{T,t}}$, so

$$\mathbb{E}_{\mathbb{P}}[G(X_t)] = Z_T^{-1}\,\mathbb{E}_0\big[G(X_t)\mathrm{e}^{-V_T(X_T)}\big] = Z_T^{-1}\,\mathbb{E}_0\big[G(X_t)\mathrm{e}^{-V_t(X_t)}\big].$$

The Polchinski equation

Theorem. (Polchinski 1984) (formal)

$$\partial_t V_t = \tfrac{1}{2}\langle D V_t, \dot{Q}_t \, D V_t\rangle - \tfrac{1}{2}\mathrm{Tr}(\dot{Q}_t \, D^2 V_t), \qquad V_T \textit{given}.$$

Proof. $u_t = \mathrm{e}^{-V_t} = \mathrm{e}^{\frac{1}{2}\Delta_{Q_T,t}} u_T$, $\Delta_Q = \mathrm{Tr}(Q\,D^2)$, solves the backward heat equation $\partial_t u_t = -\frac{1}{2}\mathrm{Tr}(\dot{Q}_t D^2 u_t)$. Insert $D^2 u = (D V \otimes D V - D^2 V)\, u$.

Nonlinear term: generates new, non-local interactions: **trees**.

Trace term: contracts pairs of fields with the shell covariance $\dot{Q}_t \, \mathrm{d}t$: **loops**.

A backward Hamilton–Jacobi equation; $u = \mathrm{e}^{-V}$ is a Cole–Hopf transform.

The Polchinski process

Theorem. (FBSDE for the interacting field) (formal) *Under* $\mathbb{P}$, *the* ***interacting field at resolution*** t $\Phi_t := X_t$ *solves*

$$\mathrm{d}\Phi_t = \dot{Q}_t F_t \,\mathrm{d}t + \dot{Q}_t^{1/2} \mathrm{d}W_t^{\mathbb{P}}, \qquad F_t = -D V_t(\Phi_t) = \mathbb{E}_t^{\mathbb{P}}\left[-D V_T(\Phi_T)\right].$$

The ***effective force*** F *is a* $\mathbb{P}$*-martingale: an observable field of the interacting theory.*

Proof. Density process $M_t = Z_T^{-1} \mathrm{e}^{-V_t(X_t)}$. Itô + Polchinski: no drift, $\mathrm{d}M_t = -M_t \langle D V_t(X_t), \dot{Q}_t^{1/2} \mathrm{d}W_t \rangle$. Girsanov: $W_t^{\mathbb{P}} = W_t + \int_0^t \dot{Q}_s^{1/2} D V_s(X_s)\,\mathrm{d}s$. Finally $D V_t(\phi) = \mu_t^{\phi}(D V_T)$.

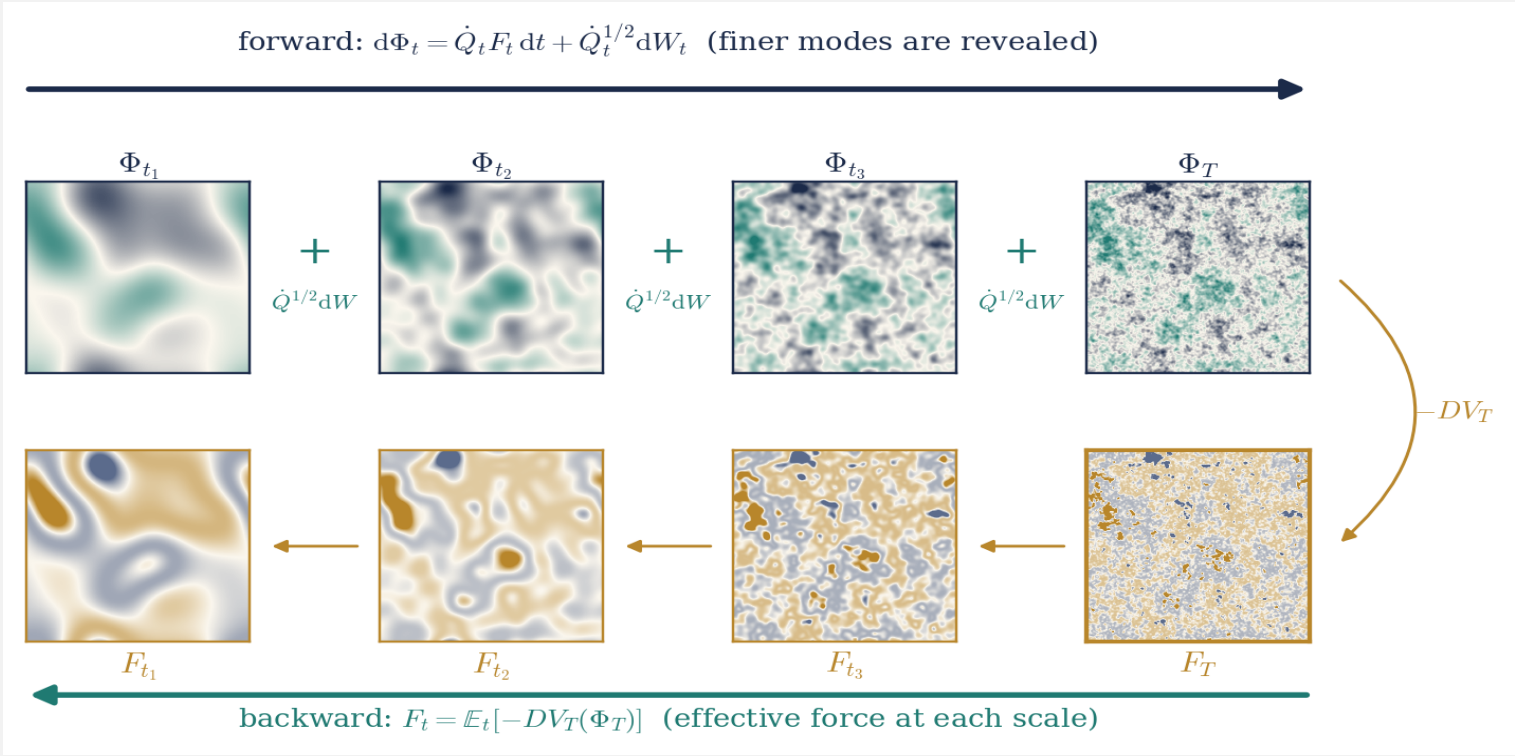
A forward–backward system

Read as an equation for an unknown pair (Φ, F):

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t = \mathbb{E}_t\left[-D\,V_T(\Phi_T)\right], \qquad t \in [0,T].$$

- The field is driven **forward** by a force determined **backward** from the terminal field.
- No Polchinski equation needed; $\nu_T = \mathrm{Law}(\Phi_T)$ from the solution.
- **Strategy**: solve with renormalised V_T, estimates uniform in T and volume, take limits.
- Φ is **not** an observable (a semimartingale); observables are built from the noise and from F.

Forward field, backward force



$\mathbb{T}^2$, heat profile, same noise. The force is shown to first order in λ: $F_t = -4\lambda H_3(\Phi_t; c_t)$, each panel in its own units.

Closed versus local

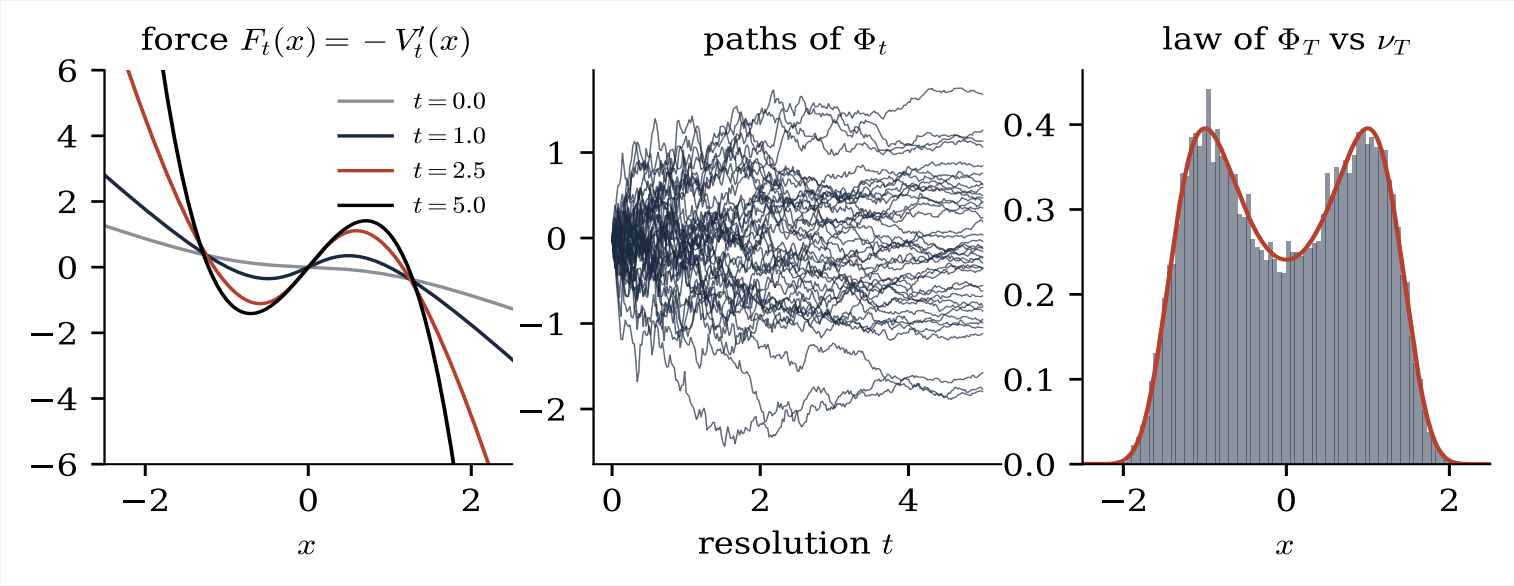
At finite T the force is a **closed** observable, $F_t = \mathbb{E}_t[F_T]$, and for Φ^4 it is **local**:

$$F_T(x) = -4\lambda H_3(\Phi_T(x); c_T) - 2\gamma_T H_1(\Phi_T(x); c_T).$$

As $T \to \infty$

F should converge **as an observable**. In $d = 2$ it is closed, $F_\infty = -4\lambda{:}\Phi^3_\infty{:}$. In $d = 3$ a closed limit $F_t = \mathbb{E}_t F_\infty$ would let the DS identity pass to $t \to \infty$, and conditioning on $\sigma(\Phi_\infty)$ would give an integrable logarithmic derivative of Φ^4_3 as a function of the field: not expected (Albeverio–Liang–Zegarlinski 2006: the regularised ones do not converge in L^2). Locality of the limiting force is the crux of the construction.

Zero dimensions



$Q_t = 1 - \mathrm{e}^{-t}$, $T = 5$, $V_T(x) = \frac{1}{2}x^4 - \frac{3}{2}x^2$. Left: $F_t = -V_t'$, almost linear at low resolution, double well as $t \to T$. Centre: FBSDE paths split between the wells (stochastic localisation). Right: histogram of Φ_T vs. the density of ν_T.

Observables of the interacting theory

For G a functional of the terminal field, $O_t = \mathbb{E}_t^{\mathbb{P}}[G(\Phi_T)] = G_t(\Phi_t)$, $G_t(\phi) = \mu_t^{\phi}(G)$, and

$$\partial_t G_t + \tfrac{1}{2}\mathsf{Tr}(\dot{Q}_t D^2 G_t) - \langle \dot{Q}_t D V_t, D G_t \rangle = 0, \qquad G_T = G:$$

the backward Kolmogorov equation, a Doob transform of the Gaussian heat equation. Any solution gives an observable, closed at finite T, not necessarily as $T \to \infty$. The force is $G = -D V_T$.

Tangent observables. V_T^{θ} depending on a coupling, $U_t = \partial_\theta V_t^{\theta}$:

$$U_t(\Phi_t) = \mathbb{E}_t^{\theta}[\partial_\theta V_T^{\theta}(\Phi_T)], \qquad \partial_\theta \mathbb{E}^{\nu_T^{\theta}}[G] = -\mathsf{Cov}^{\nu_T^{\theta}}(G, \partial_\theta V_T^{\theta}).$$

E.g. the running $:\varphi^4:$ for $\theta = \lambda$: observables **conjugate to the coordinates of theory space**.

The force: a Burgers equation

Differentiating Polchinski, $F_t = -D V_t$ solves the backward viscous **Burgers equation**

$$\partial_t F_t = -D F_t[\dot{Q}_t F_t] - \tfrac{1}{2}\mathsf{Tr}(\dot{Q}_t D^2 F_t), \qquad F_T = -D V_T.$$

- Cole–Hopf $F = D\log u$ linearises it ($u = \mathrm{e}^{-V}$ solves a heat equation).
- Uses only the force: **intensive**, while V_t is extensive ($\sim$ volume). Matters in infinite volume.
- Flows for different covariances commute when DF is symmetric, i.e. F a gradient. Non-gradient forces (general Wilson–Itô diffusions): curvature.

The FBSDE in force coordinates: the Wetterich equation

Lecture 3

State and costate

$S_t := Q_T - Q_t$: the covariance still unresolved; $\dot{S}_t = -\dot{Q}_t$, $S_T = 0$. $\mathrm{d}F_t = \Upsilon_t \,\mathrm{d}W_t$.

Pontryagin. Control problem $\mathrm{d}X = \dot{Q}_t^{1/2}(u\,\mathrm{d}t + \mathrm{d}W)$, cost $V_T(X_T) + \frac{1}{2}\int \|u\|^2$. Hamiltonian $\langle p, \dot{Q}_t^{1/2} u\rangle - \frac{1}{2}\|u\|^2$ is state independent $\Rightarrow \mathrm{d}p_t = \Upsilon_t \,\mathrm{d}W_t$, $p_T = -DV_T(X_T)$, $u_t = \dot{Q}_t^{1/2} p_t$.

The field Φ is the state; the force F is the costate.

- $F_t(\cdot)$ is the **decoupling field**, Burgers its quasilinear PDE.
- Transport term $-DF[\dot{Q}F]$ ← drift of the state: trees.
  Itô term $-\frac{1}{2}\mathrm{Tr}(\dot{Q}D^2F)$ ← noise: loops.

Noiseless flow from t to T: the linear **shear** $\Sigma_t : (x, p) \mapsto (x + S_t p, p)$, the characteristics of the transport part.

Removing the predictable drift: the forecast

Proposition. (formal) *The conditional mean of the terminal field is*

$$\psi_t := \mathbb{E}_t[\Phi_T] = \Phi_t + S_t F_t, \qquad (\psi_t, F_t) = \Sigma_t(\Phi_t, F_t),$$

a martingale with $\mathrm{d}\psi_t = \dot{Q}_t^{1/2}\mathrm{d}W_t + S_t\,\mathrm{d}F_t = (1 + S_t D F_t(\Phi_t))\,\dot{Q}_t^{1/2}\mathrm{d}W_t$.

Proof. F martingale $\Rightarrow \mathbb{E}_t \int_t^T \dot{Q}_s F_s\,\mathrm{d}s = S_t F_t$. In the differential, $\dot{Q}_t F_t\,\mathrm{d}t$ cancels with $\dot{S}_t F_t\,\mathrm{d}t$.

- $\Phi_t = \psi_t - S_t F_t$: a martingale (the **field observable**) minus the drift still expected on the unresolved scales.
- Noise of ψ: fresh shell fluctuation + **revision of the forecast**. The forward–backward coupling moves from the drift to the volatility.

ψ is the **interaction picture** w.r.t. the classical dynamics.

The FBSDE in forecast coordinates

Invert $x \mapsto \psi = x + S_t F_t(x)$: $x = \psi + S_t F_t(\psi)$, $F_t(\psi) := -F_t(x)$. Response and regulator:

$$H_t(\psi) = (S_t^{-1} + DF_t(\psi))^{-1}, \qquad R_t = S_t^{-1} - Q_T^{-1}, \qquad \dot{R}_t = S_t^{-1}\dot{Q}_t S_t^{-1} \geq 0.$$

Theorem. (formal)

$$\mathrm{d}\psi_t = H_t(\psi_t)\; S_t^{-1}\dot{Q}_t^{1/2}\,\mathrm{d}W_t, \qquad F_t(\psi_t) = \mathbb{E}_t\left[D V_T(\psi_T)\right],$$

$\psi_T = \Phi_T$. ***No drift***; *covariance* $\mathscr{C}_t = H_t \dot{R}_t H_t^*$. *The decoupling field solves the* ***pure diffusion*** *equation*

$$\partial_t F_t + \tfrac{1}{2}\mathrm{Tr}(\mathscr{C}_t\, D^2 F_t) = 0, \qquad F_T = D V_T.$$

Proof. $1 + S_t DF_t = H_t S_t^{-1}$; $F_t(\psi_t)$ is a martingale and ψ has no drift: Itô.

What the change of coordinates did

Compare with Burgers:

$$\underbrace{\partial_t F = -DF[\dot{Q}F] - \tfrac{1}{2}\mathrm{Tr}(\dot{Q}D^2F)}_{\text{field coordinate}} \quad \longrightarrow \quad \underbrace{\partial_t F = -\tfrac{1}{2}\mathrm{Tr}(\mathscr{C}\, D^2F)}_{\text{forecast coordinate}}$$

- The transport term is **absorbed exactly**: dropping the Itô term, $F_t = D\,V_T$ does not move. All **trees** live in the coordinate change $x = \psi + S_t F_t(\psi)$.
- **Loops** come only from the quadratic variation $\mathscr{C}_t$ of the forecast; the nonlinearity has moved into its field dependence.
- No gradient structure, no Gaussian convolution used: valid for any FBSDE with a Markov martingale force, whenever the coordinate change is invertible.

Conditional moments

Proposition. (formal) *If* $F_t = -D V_t$, *then*

$$H_t(\psi_t) = \mathrm{Cov}_t := \mathrm{Cov}^{\mathbb{P}}(\Phi_T \mid \mathscr{F}_t) = S_t - S_t D^2 V_t(\Phi_t)\, S_t.$$

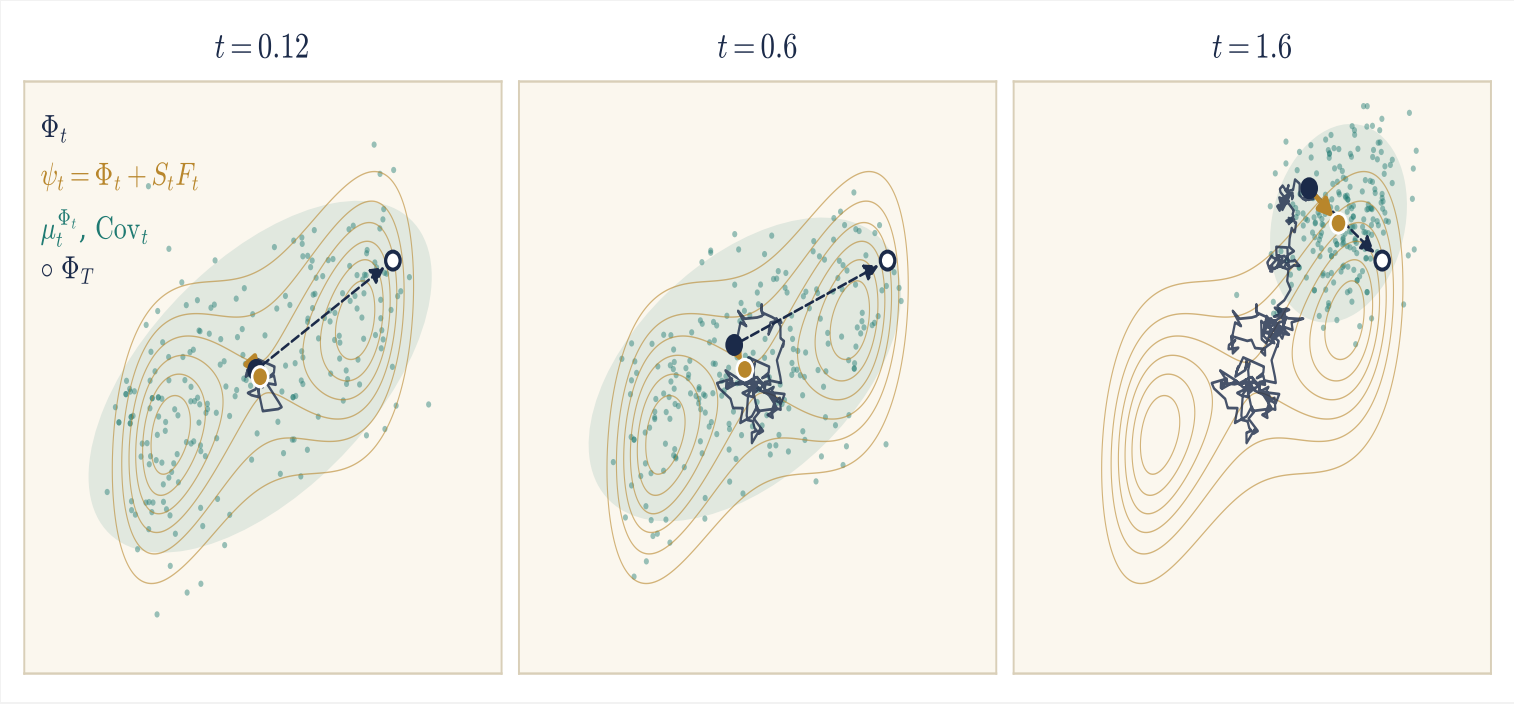
So $H_t > 0$, $1 + S_t D F_t$ *is always invertible, and* $H_t \le S_t$ *for convex* V_T *(Brascamp–Lieb).*

The forecast coordinates are the first two conditional moments of the stochastic localisation: $(\psi_t, \mathrm{Cov}_t) = (\mathbb{E}_t \Phi_T, \mathrm{Cov}_t \Phi_T)$. Since $\mathbb{E}_t[\Phi_T \otimes \Phi_T] = \psi_t \otimes \psi_t + \mathrm{Cov}_t$ is a martingale,

$$\mathrm{d}\langle \psi, \psi \rangle_t = \mathrm{Cov}_t \dot{R}_t \mathrm{Cov}_t \, \mathrm{d}t, \qquad \mathrm{dCov}_t = -\mathrm{Cov}_t \dot{R}_t \mathrm{Cov}_t \, \mathrm{d}t + \mathrm{d}(\mathrm{mart.}).$$

Gaussian reference: $S_t \dot{R}_t S_t = \dot{Q}_t$.

Stochastic localisation: a picture



Two-component field, V_T a tilted double well (contours: ν_T). Path Φ_t, forecast $\psi_t = \Phi_t + S_t F_t$, conditional law $\mu_t^{\Phi_t}$ and 2σ ellipse of Cov_t; open circle: Φ_T. The conditional law localises on one well.

The gradient sector

$F_t = -D\,V_t$: the graph $\{(x, -D\,V_t(x))\}$ is a **Lagrangian submanifold** of (field, force) space. The shear Σ_t is symplectic (S_t symmetric), and the image is a graph over ψ ($x \mapsto x + S_t F_t(x)$ is $S_t\,D$ of a strictly convex function). Hence

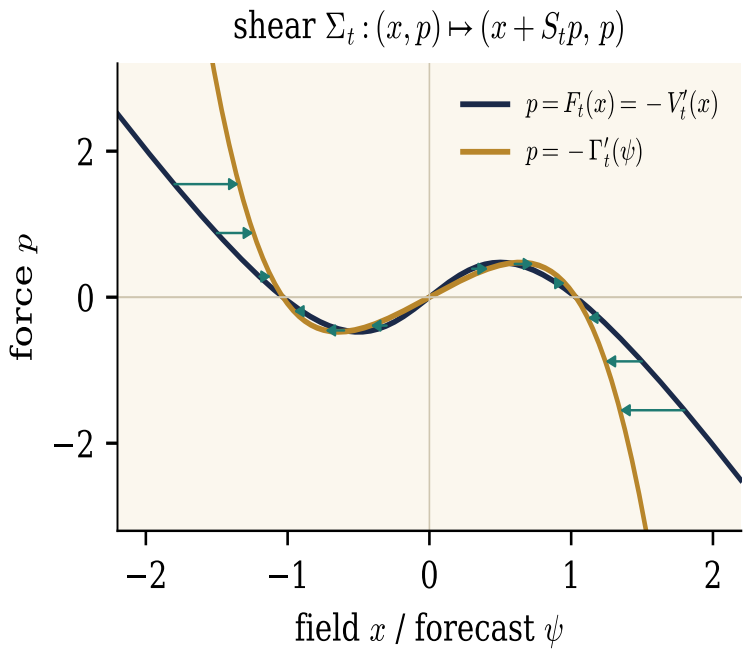
$$F_t = D\,\Gamma_t, \qquad H_t = (S_t^{-1} + D^2\Gamma_t)^{-1}, \qquad \Gamma_T = V_T.$$

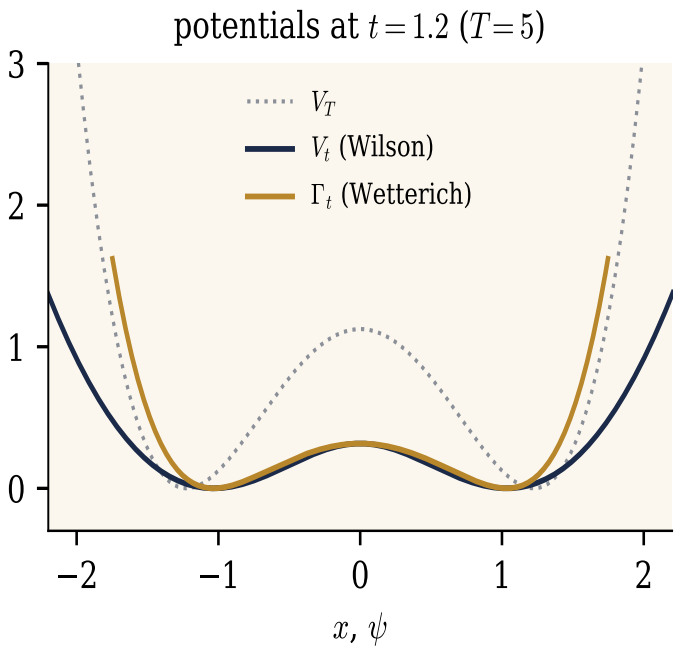
Generating function: the **Moreau transform**

$$V_t(x) = \inf_{\psi} \left\{ \Gamma_t(\psi) + \tfrac{1}{2}\langle x - \psi, S_t^{-1}\,(x - \psi)\rangle \right\},$$

optimality: $x = \psi + S_t\,D\,\Gamma_t(\psi)$, $D\,V_t(x) = D\,\Gamma_t(\psi)$. Well defined: $\frac{1}{2}\,\langle x, S_t^{-1}\,x\rangle - V_t(x)$ has Hessian $S_t^{-1}\mathrm{Cov}_t S_t^{-1} > 0$.

The shear in zero dimensions





Left: Σ_t maps the graph of $F_t = -V_t'$ to the graph of $-\Gamma_t'$ over the forecast. Right: V_T, V_t and Γ_t ($t = 1.2$, $T = 5$).

The Wetterich equation

Theorem. (formal)

$$\partial_t \Gamma_t = \tfrac{1}{2}\mathrm{Tr}\big[\dot{R}_t\,((S_t^{-1} + D^2\Gamma_t)^{-1} - S_t)\big], \qquad \Gamma_T = V_T.$$

Equivalently, for the ***effective average action*** $\Gamma_t^{\mathsf{EAA}} = \Gamma_t + \frac{1}{2}\langle \psi, Q_T^{-1}\psi\rangle$,

$$\partial_t \Gamma_t^{\mathsf{EAA}} = \tfrac{1}{2}\mathrm{Tr}\big[\dot{R}_t\,(D^2\Gamma_t^{\mathsf{EAA}} + R_t)^{-1}\big] + \textit{const},$$

from the bare action ($t = T$, $R_T = \infty$) to the 1PI effective action of ν_T ($t = 0$, $R_0 = 0$).

Proof. $D_v \mathrm{Tr}(\dot{R}H) = -\mathrm{Tr}(\mathscr{C}\, D^2 F)[v]$ by symmetry of $D^3\Gamma$, so the force equation reads $D\left(\partial_t\Gamma - \frac{1}{2}\mathrm{Tr}(\dot{R}H)\right) = 0$. The constant: envelope theorem + Polchinski.

Wetterich = the backward half of the FBSDE

Every object in the Wetterich equation is an observable of the Polchinski process.

Wetterich	FBSDE
field argument ψ	forecast $\psi_t = \mathbb{E}_t[\Phi_T]$
$D\Gamma_t(\psi_t)$	minus the costate, $\mathbb{E}_t[D V_T(\Phi_T)]$
propagator $(D^2\Gamma_t^{\mathsf{EAA}} + R_t)^{-1}$	conditional covariance Cov_t
regulator $R_t = S_t^{-1} - Q_T^{-1}$	precision of the unresolved shell minus the prior's
$\partial_t \Gamma_t$	$\frac{1}{2}\mathrm{Tr}[\dot{R}_t\,(\mathsf{Cov}_t - S_t)]$

Driven by **interacting minus free** conditional covariance; **one-loop exact**, with the full propagator; in the Gaussian case only the constant flows. Wetterich 1993; Morris 1994; Berges–Tetradis–Wetterich 2002

The regulator of the heat profile

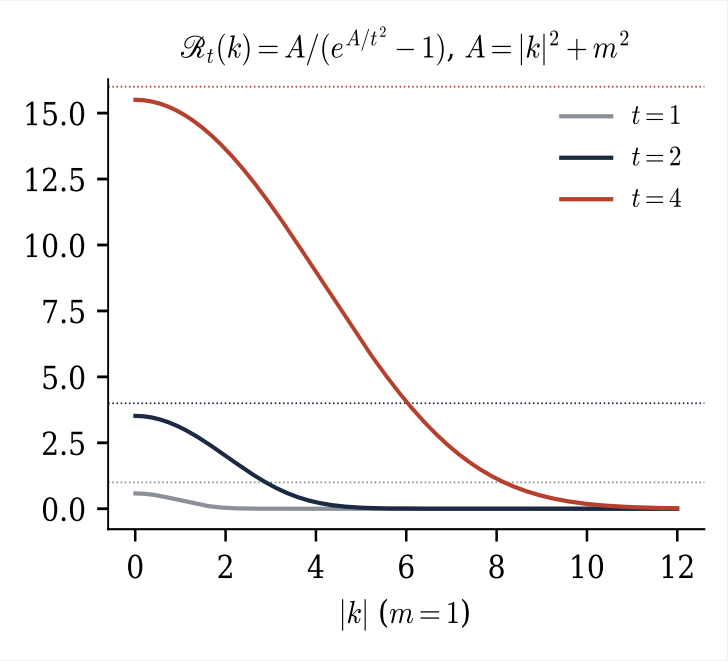
For $T=\infty$, $S_t = C\,(1-\mathrm{e}^{-A/t^2})$ and

$$R_t = \frac{A}{\mathrm{e}^{A/t^2}-1}.$$

- $\approx t^2$ (a mass term) on the modes $|k| \ll t$;
- exponentially small on $|k| \gg t$.

The **exponential regulator** of the functional RG.

Other profiles give other regulators, e.g. the mass regulator $R_t = t^2$.



Variational meaning and the tree level

Constrained free energy. Gibbs variational principle for the conditional law:

$$\Gamma_t(\psi) = \inf\left\{\int V_T \,\mathrm{d}\nu + \mathsf{Ent}(\nu \,|\, \mu^{S_t}(\cdot - \psi)) \colon \int \varphi \,\mathrm{d}\nu = \psi\right\}\colon$$

the free energy of the unresolved shell, **constrained to the forecast**. V_t is the value function; Γ_t is its Moreau transform.

Tree level. Drop the Itô term: $\Gamma_t = V_T$, and the Moreau transform becomes the **Hopf–Lax formula**

$$V_t^{\mathsf{tree}}(x) = \inf_y \left\{V_T(y) + \tfrac{1}{2}\langle y - x, S_t^{-1}(y-x)\rangle\right\},$$

which solves $\partial_t V = \frac{1}{2}\langle DV, \dot{Q} DV\rangle$: the saddle point of the Gaussian convolution. The minimiser solves the classical field equation $S_t^{-1}(y-x) + DV_T(y) = 0$ in the shell $[t,T]$; expanding in V_T gives exactly the tree diagrams.

Variational structure

Lecture 3

The Boué–Dupuis formula

$\mathbb{H}_a$: adapted drifts with $\mathbb{E}\int_0^T \|u_s\|^2 < \infty$; $X^u = X + \int_0^\cdot \dot{Q}_s^{1/2} u_s \, \mathrm{d}s$.

Theorem. (Boué–Dupuis 1998; Polchinski = HJB) (formal)

$$-\log \mathbb{E}_0[\mathrm{e}^{-V_T(X_T)}] = V_0(0) = \inf_{u \in \mathbb{H}_a} \mathbb{E}_0\left[V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_s\|^2 \mathrm{d}s \right].$$

Optimal feedback $u_s^* = -\dot{Q}_s^{1/2} D V_s(X_s^{u^*})$, $\mathrm{Law}(X_T^{u^*}) = \nu_T$. *$V_t$ is the* ***value function****; Polchinski is the* ***HJB equation****.*

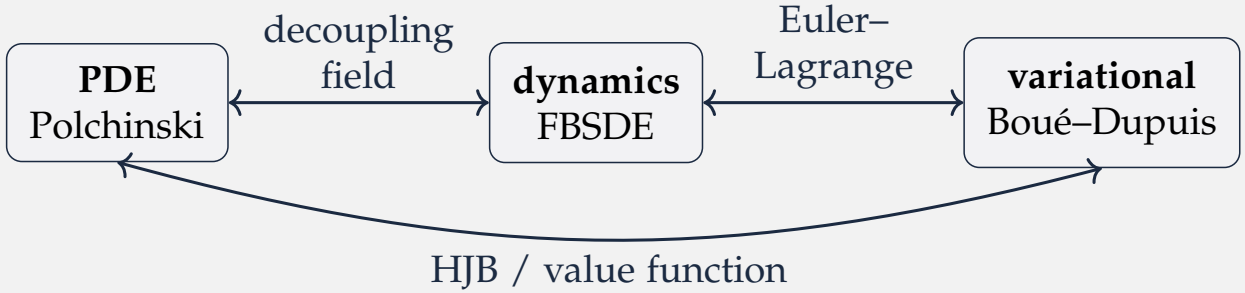
Proof (verification). Itô + Polchinski:

$$\mathbb{E}_0\left[V_T(X_T^u) + \tfrac{1}{2}\int_0^T \|u_t\|^2 \right] = V_0(0) + \tfrac{1}{2}\mathbb{E}_0 \int_0^T \|u_t + \dot{Q}_t^{1/2} D V_t(X_t^u)\|^2 \mathrm{d}t.$$

Three faces of one object

Euler–Lagrange. Perturb $u \to u + \varepsilon v$:

$$u_s = -\dot{Q}_s^{1/2}\, \mathbb{E}_s\big[D V_T(X_T^u)\big] = \dot{Q}_s^{1/2} F_s: \quad \text{the FBSDE.}$$



Entropy. $\frac{1}{2}\,\mathbb{E}_0 \int \|u\|^2 \geq \mathsf{Ent}(\mathsf{Law}(X^u)|\mathsf{Law}(X))$: a pathwise lift of the Gibbs principle $-\log Z_T = \inf_\nu \{\int V_T \,\mathrm{d}\nu + \mathsf{Ent}(\nu|\mu^{Q_T})\}$. Useful because adapted drifts can be **built scale by scale**, as functionals of X.

Renormalisation in the variational picture

Generating functional:

$$\mathscr{W}_T(f) = -\log \int \mathrm{e}^{-f}\,\mathrm{d}\nu_T = \inf_u \mathbb{E}_0\Big[(f+V_T)(X_T^u) + \frac{1}{2}\int_0^T \|u_s\|^2\Big] - V_0(0).$$

Bounds uniform in $T \Rightarrow$ tightness, moments, identification of limits.

Φ^4_3. The leading drift $u_s \approx -4\lambda\,\dot{Q}_s^{1/2} {:}X_s^3{:}$ has **divergent cost**,

$$\tfrac{1}{2}\mathbb{E}\int_0^T \|4\lambda\,\dot{Q}_s^{1/2} {:}X_s^3{:}\|^2 \mathrm{d}s = 12\lambda^2 L^3 \int Q_T(z)^4\,\mathrm{d}z \sim T.$$

Change variables $u = -4\lambda\,\dot{Q}^{1/2} {:}X^3{:} + r$, then paracontrolled changes of variables: the singular terms become explicit polynomials in Wick powers of X, **cancelled by the counterterms**, and the remainder is coercive. Uniform control in Besov/paracontrolled norms = construction of Φ^4_3. Barashkov–Gubinelli, Duke 2020; EJP 2021

Infinite volume

On $\mathbb{T}_L^d$, $V_0(0)$ and V_t are **extensive** ($\sim L^d$). Three **intensive** objects survive:

i. **the force.** For $V_T = \int v_T(\varphi)$, on $\mathbb{R}^d$:

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t(x) = -\mathbb{E}_t\big[v_T'(\Phi_T(x))\big].$$

Everything is local; infinite-volume limits should solve this. Uniqueness of solutions $\leftrightarrow$ uniqueness of the theory; non-uniqueness expected at phase transitions (Φ_2^4, deep double well).

ii. **the Burgers flow** of the force;

iii. **differences of free energies** $\mathscr{W}_T^{(L)}(f)$, f local: bounded as $L \to \infty$; basis of the infinite-volume analysis of Φ_2^4 and $\exp(\Phi)_2$.

Estimates must be **local** (weighted norms, volume-independent constants): the analogue of "coming down from infinity". Barashkov–Gubinelli, PMP 2023; Glimm–Jaffe–Spencer 1975

Additional material

Lecture 3

Wick observables in every dimension

Recall $\partial_c H_n(a;c) = -\frac{1}{2}\partial_a^2 H_n$, $\partial_a H_n = n H_{n-1}$.

Proposition. *For every d, n, x, $:X_t^n:(x) := H_n(X_t(x);c_t)$ is a continuous martingale,* $\mathrm{d}:X_t^n:(x) = n:X_t^{n-1}:(x)\,\mathrm{d}X_t(x)$, *and*

$$\mathbb{E}\left[:X_t^n:(x):X_t^k:(y)\right] = \delta_{nk}\, n!\, Q_t(x,y)^n \xrightarrow{t\to\infty} \delta_{nk}\, n!\, G(x-y)^n \quad (x \neq y).$$

Proof. Itô: the drift is $(\frac{1}{2}\partial_a^2 + \partial_c) H_n\, \dot{c}_t = 0$.

- Lecture 2: $:\varphi^n:$ exists iff $n(d-2) < d$. As **observables**, Wick powers exist in every d, unsmeared.
- For $n(d-2) \geq d$, $0 \leq f \not\equiv 0$, $:X_t^n:(f)$ is unbounded in L^2 and (Paley–Zygmund) not tight: **no terminal value**; the observable is the whole family.
- The singularity has moved to **products at coincident points**.

Products of observables

Definition. *$O^1 \star O^2$ is the observable with $\|(O^1 \star O^2)_t - O^1_t\, O^2_t\|_{L^1} \to 0$; unique, and $(O^1 \star O^2)_s = \lim_t \mathbb{E}_s[O^1_t\, O^2_t]$.*

Proposition. *If O^1, O^2 are continuous, square integrable on bounded intervals, and $\mathbb{E}\int_0^\infty |\mathrm{d}\langle O^1, O^2\rangle_s| < \infty$, then*

$$(O^1 \star O^2)_t = O^1_t\, O^2_t + \mathbb{E}_t\Big[\int_t^\infty \mathrm{d}\langle O^1, O^2\rangle_s\Big].$$

Proof. $O^1\, O^2 - \langle O^1, O^2\rangle$ and $\mathbb{E}_t\langle O^1, O^2\rangle_\infty$ are martingales.

Products of point evaluations

$\langle \mathbf{X}(x), \mathbf{X}(y) \rangle_t = Q_t(x-y)$ is deterministic. Heat profile, $d \geq 2$: the condition holds iff $x \neq y$, and then

$$(\mathbf{X}(x) \star \mathbf{X}(y))_t = X_t(x)\, X_t(y) + G(x-y) - Q_t(x-y), \qquad \langle \mathbf{X}(x) \star \mathbf{X}(y) \rangle = G(x-y).$$

The correct two-point function, **without ever discussing the regularity of X_∞**. For $x = y$: no product, since $c_t \to \infty$.

OPE. At every fixed resolution, as $y \to x$,

$$(\mathbf{X}(x) \star \mathbf{X}(y))_t - G(x-y) \longrightarrow X_t(x)^2 - c_t =: X_t^2 : (x),$$

i.e. $\mathbf{X}(x) \star \mathbf{X}(y) = G(x-y)\, 1 + : \mathbf{X}^2 : (x) + o(1)$: the Wick square is the regular part of the product.

The operator product expansion

For $x \neq y$,

$$:\mathbf{X}^n:(x) \star :\mathbf{X}^k:(y) = \sum_{j \leq n \wedge k} j! \binom{n}{j} \binom{k}{j} G(x-y)^j :\mathbf{X}^{n-j}(x)\mathbf{X}^{k-j}(y):.$$

- Wick's theorem at the level of observables.
- The **coefficients** $G(x-y)^j$ carry all singularities; the observables on the right have regular limits as $y \to x$.

Locality and factorisation

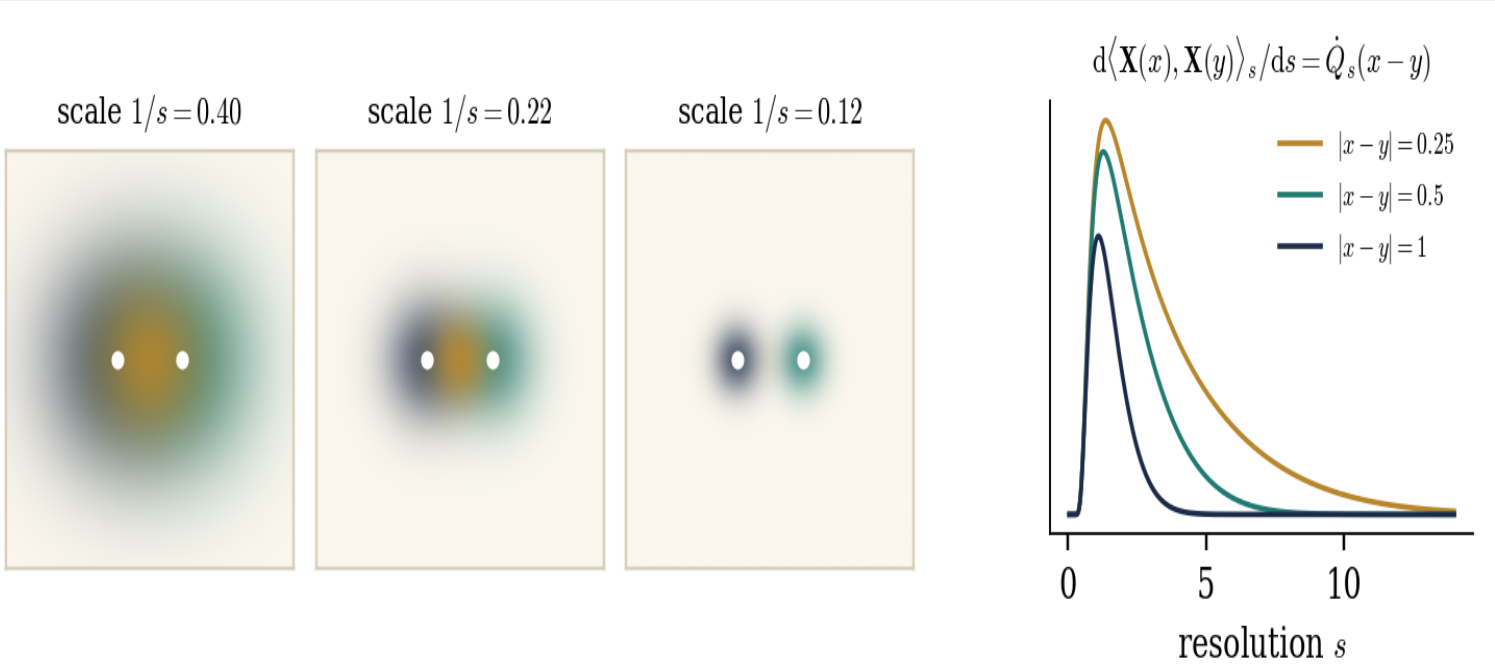
- O^1, O^2 with germs supported in $U^1_{1/t}$, $U^2_{1/t}$, $\mathrm{dist}(U^1, U^2) > 0$: the covariation $\langle D O^1_t, \dot{Q}_t \, D O^2_t \rangle \mathrm{d}t$ sees $\dot{Q}_t$ only between the supports.
- Finite range decomposition: it vanishes for large t, so the product exists and is asymptotically the pointwise product. Heat profile: exponentially small.
- Observables on **disjoint regions** multiply freely; products at the **same point** need renormalisation.

Factorisation algebra

This is the structure of a **factorisation algebra** of observables (precise axioms not established here).

Bauerschmidt–Brydges–Slade, LNM 2242 (finite range decompositions)

Locality: a picture



Left: kernels of $\dot{Q}_s^{1/2}$ at x and y; their overlap integrates to $\dot{Q}_s(x-y) = \mathrm{d}\langle \mathbf{X}(x), \mathbf{X}(y)\rangle_s / \mathrm{d}s$. Right: negligible for $s \gg 1/|x-y|$; total $G(x-y)$. As $y \to x$ the mass escapes to $s \to \infty$.

A semigroup of renormalisation maps

$\mathcal{R}_S(V) = -\log\,(\mu^S * \mathrm{e}^{-V})$. Since $\mu^S * \mu^{S'} = \mu^{S+S'}$,

$$\mathcal{R}_S \circ \mathcal{R}_{S'} = \mathcal{R}_{S+S'} = \mathcal{R}_{S'} \circ \mathcal{R}_S, \qquad V_t = \mathcal{R}_{Q_{T,t}}(V_T):$$

an orbit of an **abelian** semigroup indexed by the cone of positive covariances, non-linear on V, linear on e^{-V}. Generators:

$$\mathcal{P}_B(V) = \tfrac{1}{2}\mathrm{Tr}(B\,D^2 V) - \tfrac{1}{2}\langle D V, B\,D V\rangle, \qquad \partial_t V_t = -\mathcal{P}_{\dot{Q}_t}(V_t).$$

Proposition. (Flatness) (formal) $[\mathcal{P}_B, \mathcal{P}_{B'}] = 0$. *$V_t$ depends on the profile only through $Q_{T,t}$; $V_0(0)$ and ν_T only through Q_T.*

Different profiles with the same endpoints are different **coordinate systems** on the same RG trajectory: they change Φ and intermediate observables, not the terminal theory. **No holonomy.**

Perturbation theory and counterterms

Gaussian convolution **lowers the Wick ordering**: $V_t = \int [\lambda{:}\phi^4{:}_t + \gamma_T {:}\phi^2{:}_t] + O(\lambda^2)$, the dual of $\int H_4(X_t; c_t)$ being an observable. At second order, $S = Q_{T,t}$:

$$V_t = \int [\lambda{:}\phi^4{:}_t + \gamma_T {:}\phi^2{:}_t] - \frac{\lambda^2}{2} \sum_{k=1}^{4} \binom{4}{k}^2 k! \iint S(x,y)^k {:}\phi^{4-k}{:}_t(x) {:}\phi^{4-k}{:}_t(y) + O(\lambda^3).$$

$T \to \infty$ at fixed t, $d = 3$, $G(z) \sim (4\pi|z|)^{-1}$:

- $k = 1, 2$: finite, nonlocal.
- $k = 3$ (**sunset**): $S^3 \sim |z|^{-3}$; local divergence $-48\lambda^2 (\int S^3) \int {:}\phi^2{:}_t$ (+ const), cancelled by $\gamma_T = 48\lambda^2 \int Q_T^3 = \frac{3\lambda^2}{\pi^2} \log T + O(1)$.
- $k = 4$: vacuum energy $\sim T$; cancels in ν_T.

Renormalisation = choosing V_T so that V_t at **fixed** resolution converges. $\nu_t \ll \mu^{Q_t}$ expected for $t < \infty$; the terminal Φ^4_3 is singular w.r.t. the GFF.

Dyson–Schwinger equations at every scale

Operators functions of A; $K = Q_T^{-1}$, $J_t = K Q_t$ (heat profile: a heat semigroup, $\to e^{-A/t^2}$).

Theorem. (formal) *The **noise observable** $\eta_t = K\Phi_t - J_t F_t$ is a centred martingale,* $d\eta_t = K \dot{Q}_t^{1/2} dW_t^{\mathbb{P}} - J_t dF_t$, *and*

$$\mathbb{E}^{\mathbb{P}}\left[D_{J_t h} G(\Phi_t)\right] = \mathbb{E}^{\mathbb{P}}\left[G(\Phi_t)\,\langle h, K\Phi_t - J_t F_t\rangle\right].$$

At $t = T$: the DS equation $\mathbb{E}^{\nu_T}[D_h G] = \mathbb{E}^{\nu_T}[G\langle h, K\varphi + D V_T\rangle]$.

Proof. $\dot{J}_t = K\dot{Q}_t$: drifts cancel. IBP for μ^{Q_t} in direction $J_t h$, using $\Phi_t \sim \nu_t$.

The field equation at resolution t

As $T \to \infty$, $K \to A$, and $\eta_t = K\Phi_t - J_t F_t$ becomes

$$A\Phi_t = A Q_t F_t + \eta_t \qquad (A Q_t = \mathrm{e}^{-A/t^2} \text{ for the heat profile}):$$

the Euclidean field equation $A\varphi = -V'(\varphi) + \text{noise}$, **seen at resolution t**. No products of distributions: the nonlinearity enters only through the force **observable**, smeared at scale $1/t$. For Φ^4_3, with F_t in the role of the renormalised cube, $t \to \infty$ corresponds to the renormalised DS equation (Gubinelli–Hofmanová 2021).

Compare: stochastic quantisation (Lecture 2)

There the field equation is imposed in an **auxiliary time**; here it holds at **each resolution**, and the auxiliary parameter is the resolution itself.

Conditioning an SPDE solution on a scale filtration gives identities of the same shape but with a different noise: the martingale $J_t \,\mathrm{d}F_t$ in $\mathrm{d}\eta_t$ is essential.

Summary

- The GFF is the terminal value of a martingale X_t of smooth fields; **observables are martingales**, possibly without terminal value.
- Interactions: effective potentials V_t solve **Polchinski**; the interacting field solves an **FBSDE** whose force is an observable.
- In the forecast coordinate $\psi_t = \mathbb{E}_t \Phi_T$ the FBSDE has no drift, trees are integrated exactly, and the force flow is pure diffusion; in the gradient case it is **Wetterich's** equation.
- PDE (Polchinski) $\leftrightarrow$ dynamics (FBSDE) $\leftrightarrow$ variational principle (Boué–Dupuis).
- Products, OPE, counterterms and DS equations all live at **finite resolution**.

Next

Later lectures (not yet settled): probably a construction of Φ^4_3 on the torus via the FBSDE.

Exercises and reading

Exercises (numbering of the notes)

- 4.1 The field at resolution t; 4.2 divergence of c_t.
- 5.1 Quadratic interactions; 5.2 zero dimensions.
- 6.1 Stochastic maximum principle; 6.2 characteristics.
- 6.5 The constant; 6.7 1PI effective action.
- 6.8 Trees versus loops; 6.9 regulators.
- 7.2 Cost of the leading drift.
- 8.2 Products of Wick observables; 8.3 exponentials.
- 9.2 The sunset; 9.3 power counting.
- 10.1 DS from the dynamics.

Reading

- Bauerschmidt–Bodineau–Dagallier, *Stochastic dynamics and the Polchinski equation*, Probab. Surveys 2024.
- Barashkov–Gubinelli, *A variational method for* Φ^4_3, Duke 2020.
- Berges–Tetradis–Wetterich, Phys. Rep. 2002.
- Ma–Yong, *FBSDEs and their applications*, LNM 1702.
- Bailleul–Chevyrev–Gubinelli, *Stochastic observables and Wilson–Itô diffusions*, in prep.